
“Wheezer” – Disease (Laryngeal Paralysis)



Dirsko J.F. von Pfeil, Dr.med.vet, DVM

Diplomate American College of Veterinary Surgeons
Diplomate European College of Veterinary Surgeons
Adjunct Professor, Small Animal Surgery,
Michigan State University

Staff Surgeon at:

Veterinary Specialists of Alaska, P.C.
3330 Fairbanks Street
Anchorage, Alaska 99503, U.S.A.
Phone: +1 (907) 274 – 0645
Fax: +1 (907) 929 – 3320
E-mail: drvonpfeil.vsoa@ak.net



Dr. Bryden J. Stanley, BVMS, MACVSc, MVetSc,

Diplomate American College of Veterinary Surgeons
Assistant Professor, Small Animal Surgery,
Michigan State University
East Lansing, MI 48824, U.S.A.
Ph: +1 (517) 432 - 9902
Email: stanle32@cvm.msu.edu

Laryngeal Paralysis (“Wheezer-Disease”)

Dirsko J.F. von Pfeil¹, Dr.med.vet, DVM, DAVCS, DECVS, Bryden J. Stanley², BVMS, MVetSc, DACVS

¹Veterinary Specialists of Alaska, P.C. 3330 Fairbanks Street, Anchorage, AK 99503, U.S.A.

²Department of Small Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, East Lansing, MI 48824, U.S.A.

Laryngeal paralysis (LarPar) is a dysfunction of the nerves which control opening and closing of the larynx (voicebox). With this disorder, the upper airway fails to open adequately during inhalation. LarPar occurs in the Alaskan Husky. These dogs are frequently called “Wheezers” because of the typical breathing sound. The causes, signs, and treatment of LarPar are described in this chapter.

GENERAL INFORMATION AND CLINICAL SIGNS

The larynx is located at the beginning of the trachea (wind pipe). It controls airflow, voice production, enables effective coughing, and facilitates increased abdominal pressure during whelping and defecation. Specific nerves control the muscles of the larynx. With LarPar, the larynx cannot be opened or closed appropriately, secondary to nerve dysfunction. This results in inadequate oxygen supply and development of respiratory distress. It may predispose affected animals to aspiration pneumonia.^{1,2}

Clinical signs include exercise intolerance, stridor, cough, hoarse bark, choking/gagging, inspiratory dyspnea (difficulties breathing), cyanosis (blue mucous membranes), and sometimes collapse.

LarPar can be congenital (the animal is born with it), inherited (genetic defect, transmitted from one or both parents), or acquired. The acquired form can be due to trauma to the nerves (bite wounds to the neck), surgical disruption of nerves (cardiac or tracheal surgery), secondary to polyneuropathy (generalized nerve disorder), or idiopathic (no identified cause). Alaskan Huskies can be born with LarPar (“Wheezers”). It is currently suspected that this type of LarPar is an inherited condition, but a study evaluating a genetic defect in this breed could never be completed.⁴ Most affected dogs are described to have a black and/or white hair coat, “freckles”, blue eyes, and hard oral mucosal tags (Figures 1,2A,2B).



Figures 1,2A,2B: Typical appearance of Alaskan Huskies, possibly carrying the gene for LarPar (“Wheezer”): Black and/or white hair coat, “freckles”, blue eyes, and hard oral mucosal tags.

Severely affected “Wheezers” are reported to “run about 100 yards, then collapse gasping and turning blue”.⁴ They may be more prone to overheating.⁴ Apparently, some “Wheezers” compensate with age. Indeed, some Iditarod finishers were “Wheezers” at a young age.⁵ The reason for improvement cannot currently be explained. Conversely, some affected puppies require surgery to survive. Thus, the term “one-year-death-sentence” is occasionally heard.⁶ There is definitely a need for more scientific investigation focusing on LarPar in Alaskan Huskies.*

An inherited form of LarPar is described in Bouvier des Flandres, Siberian Husky-Alaskan malamute crosses, Bull Terriers, and white-coated German Shepherds.^{1,2,3} A developmental LarPar-polyneuropathy complex is described in Dalmatians, Rottweilers, Pyrenean mountain dogs, and Leonbergers.³ An acquired form affects mostly older Labradors or other large breeds.^{1,2}

* The authors are currently (2010) developing a protocol to collect DNA and pedigrees from affected dogs. Mushers with “Wheezers” are encouraged to contact Dr. von Pfeil or their regular veterinarian.

DIAGNOSIS

Diagnostics include complete physical (including auscultation and palpation of larynx and trachea) and neurological examinations. Blood work (+/- thyroid profile; Ach-Receptor-antibody test for myasthenia gravis, a neuromuscular dysfunction sometimes associated with LarPar and megaesophagus), and radiographs of chest and neck are recommended.^{1,2} Evaluation of swallowing using contrast fluoroscopy helps to determine esophageal function and may rule out hiatal herniation secondary to LarPar. Electrodiagnostic assessment of muscles and nerves are other available tests. The definitive diagnosis of LarPar is made under light anesthesia and direct visualization of the larynx. The affected larynx does not open normally during inhalation (Figure 3).



Figure 3: View of the laryngeal inlet in a dog. Normal movement of the laryngeal inlet while breathing is depicted by the white arrows and black dotted line. In dogs with LarPar, there is no such movement. Compare with Figure 5.

TREATMENT

Treatment for LarPar depends on the symptoms as the disease progresses. Thus, mildly affected animals have to be monitored carefully to avoid an emergency situation later. In very mild cases, decreasing body weight, limiting activity level, and preventing exposure to high temperatures may be sufficient. In moderate cases, mild sedatives help (Acepromazine). In severe cases, the animal is brought to a cool, non-stressful environment. Water or alcohol may be poured over the body, icepacks placed in axilla/groin for cooling. Sedation and oxygen delivery provide the most immediate relief. Occasionally, a temporary tracheostomy may be necessary. The most commonly performed surgical treatment is arytenoid lateralization (aka: "Laryngeal Tieback"). Other techniques are associated with increased complications, such as web formation later in life, obstructing airflow and leading to recurrence of symptoms. Arytenoid lateralization is technically demanding, especially when performed in a puppy, and must be performed correctly to assure acceptable outcome. (Figures 4,5).^{1,2}



Figure 4: Model of the larynx. Placement of sutures (dashed lines) during "Laryngeal Tieback", resulting in an opened airway (arrow).



Figure 5: View of the laryngeal inlet after "Laryngeal Tieback". Note the opened airway in comparison with Figure 3.

POSTOPERATIVE CARE

Surgery is usually associated with significant, immediate improvement of breathing and thus quality of life. After surgery, occasional coughing may be seen. This may improve over time if associated with irritation of the tracheal mucosa from intubation/surgery. As the larynx is permanently opened after surgery, food or water can accidentally enter the airway, and aspiration pneumonia may develop. This may be prevented by changing feeding habits and preventing swimming after surgery.

Aspiration pneumonia was evident in 18.5% of dogs with acquired idiopathic LarPar after arytenoid lateralization in a recent study.³ All dogs with pneumonia had more severe esophageal dysfunction compared to dogs that did not develop pneumonia. All patients recovered well with treatment, and no patient died from pneumonia. Interestingly, all dogs in that study developed some degree of neurological dysfunction within one year.³ Feeding from an elevated position, and keeping the animal on the hind legs for some minutes after eating is recommended. This may reduce the risk of aspiration pneumonia.³ It appears that Alaskan Huskies, operated early on in life, have a normal life expectancy and do not show the signs of polyneuropathy, seen in older dogs with acquired idiopathic LarPar. It is not known if affected Alaskan Huskies are more susceptible to aspiration pneumonia, nor has their esophageal function been evaluated.

CONCLUSION

In many dogs, LarPar appears to be a complex dysfunction not only of nerves supplying the larynx, but also of other nerves. Intensive research is performed to further investigate this potentially generalized neuromuscular disorder. The cause for LarPar in Alaskan Huskies is unknown. Although a genetic trait has not been proven yet, "Wheezers", even if clinically normal at a later stage in life, should not be used for breeding. Instead, they should be neutered in an effort to eliminate the disease from the bloodline. Surgical treatment is recommended as it usually provides fast and long standing improved quality of life.

Acknowledgments: The authors thank Dr. Mike Edwards, DVM, MS, DACVS and Cynthia White for editorial help.

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Injuries to the Axial Skeleton of Sled-Dogs



Dirsko J.F. von Pfeil, Dr.med.vet, DVM

Diplomate American College of Veterinary Surgeons
Diplomate European College of Veterinary Surgeons
Adjunct Professor, Small Animal Surgery,
Michigan State University

Staff Surgeon at:

Veterinary Specialists of Alaska, P.C.
3330 Fairbanks Street
Anchorage, Alaska 99503, U.S.A.
Phone: +1 (907) 274 – 0645
Fax: +1 (907) 929 – 3320
E-mail: drvonpfeil.vsoa@ak.net



Caroline Griffiths, DVM

President, Int. Sled Dog Veterinary Medical Association
The Traveling Vet LLC
6816 W C R 20
Loveland, CO 80537, U.S.A.
Phone: +1 (970) 613 - 0639
E-mail: chgdvm@earthlink.net

around the dog's body when pulling. The forces when pulling should be evenly distributed among all straps. There should be no bulges, wrinkles, sudden angles in the straps, and points of pressure on the skin or bony prominences when the harness is properly applied. Occasionally it may be necessary to add custom designed buffer material to the straps to prevent pressure sores (Figure 3). The use of sliding O-rings on the tug line may help to adjust for different body lengths and may decrease shock to slower starting dogs (Figure 4).

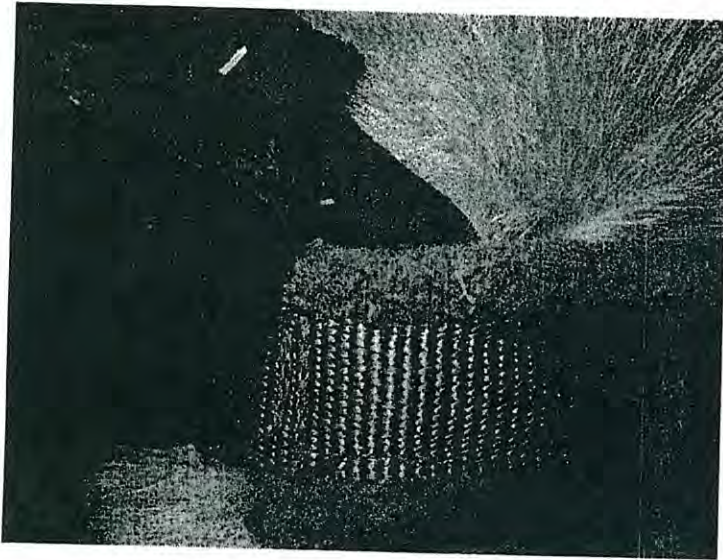
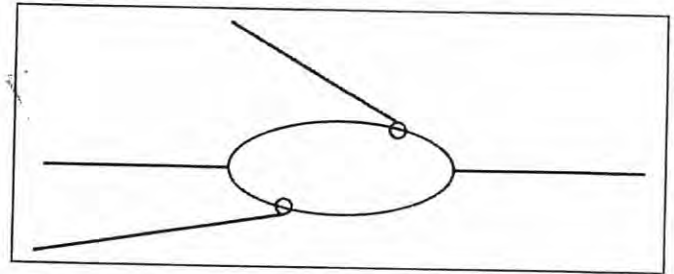


Figure 3 (left): Insulation to prevent skin sores.

Figure 4 (below): Tug line with O-rings.



Most dogs vector away from the gangline when pulling. This results in a force that is not parallel to the tow line but at a slight angle to it. Using parallel bars that are attached to the harness behind the dog at a distance long enough to not interfere with full range of motion of the hind legs, may prevent uneven pull. Especially if used without necklines, the dogs may travel in a straighter path, resulting in more even stress on the limbs.

It may be necessary to change harness types occasionally, depending on the terrain the team is travelling on. This may help to change pressure points and relieve stress on specific areas. Finally, rotating the position of the dogs may help to prevent monotonous stress on one or the other side of the dog's musculoskeletal system, thus preventing overuse injuries. Vectoring of the dogs away from the gangline while pulling creates increased stress on the outside (farthest away from the line) forelimb and inside hindlimb.

Conditioning of sporting dogs, including sled dogs, plays a major part in prevention of injuries.⁵ There is no scientific information describing the best training method or conditioning protocol for sled dogs or any sporting dogs. Therefore it is important to pay close attention to every dog during training. Screen the team for any metabolic or orthopedic diseases regularly during training and racing, and provide adequate rest to the canine athletes between runs. Conditioning starts at a young age. Animals should be evaluated for evidence of developmental orthopedic diseases between 4-9 months of age (hip dysplasia, osteochondrosis dissecans and others). Playful activities are allowed and encouraged until growth plate closure occurs, which is usually around 12 months of age. Until that time, the dog can be made familiar with the equipment. Light weights may be pulled for short intervals. Exposure to excessive stress in younger dogs may lead to fractures of the immature bone or to the growth plates. At approximately one year of age, more intense training can be initiated. The effect of conditioning is the adaptation of different body systems to the increased work, such as the gastrointestinal tract (gastric emptying, absorption of nutrients), hormonal systems (thyroid hormones), body fat metabolism, and the musculoskeletal system. Conditioning influences the musculoskeletal system by improving strength, endurance, and conscious proprioception. Strength is needed mainly in speed racing, while endurance is needed in long-distance races. The ratio of muscle mass to body weight is higher in dogs, trained for speed racing, compared to dogs trained for endurance. Training protocols and nutrition should be adjusted to reach the specific goals. It is interesting that there is no difference in performance between male and female dogs, despite differences in muscle mass and body size. Training affects all musculoskeletal tissues, and makes cartilage and ligaments stiffer, -and bones, muscles, and tendons stronger. If balance and conscious proprioception are well developed, the athlete will be able to adjust better to the challenges of different terrains during the race. Conscious proprioception can be improved by specific exercises, such as walking in specific patterns, on various flooring, over obstacles, or on grounds with different, fast changing slopes. Other exercises include rapid changes of direction, the use of cavaletti rails, exercise balls, ball playing, and walking on front or back legs,

Efficiency of all harnesses depends on proper fit. The musher must remember that the dog's body is constantly changing with weight, muscle tone, and hair coat so the fit must be monitored constantly even during races. In general, short harness types may decrease the risk of back injuries (Figure 1). These types of harnesses *may* decrease the risk of back injuries, but not other types of injuries. The authors have still seen lumbar pain in dogs with short harnesses – typically more cranial lumbar pain, versus the caudal lumbar pain that is more common with the standard harness types. Short harnesses do best without necklines. In fact, much of the benefit may be from not using necklines, rather than directly from the short harness. There is a newer type of harness being used which has better padding than the Man-Mat skijoring harness that has been used commonly in the past (Figure 1). The newer harness has rings to attach the tug-line on the side of the dog. On the other hand there may be less pulling power, because the range of motion of the hind legs is decreased. Most mushers experiment during training to find the best fit for every single dog. Some dogs may benefit from rotating different harness styles because of structure and/or gait. It is an advantage during a long race to have alternate sizes and styles available at some checkpoints for this reason.

H-Back	
Advantages	Disadvantages
Allows rib cage expansion	Neck straps may accelerate muscle fatigue especially if poor fit
Tug line angle reduces pressure on lumbosacral area, especially for long distances at a trot	Dog may be able to escape from harness
Elastic band strap alleviates pressure on the spine	Neck straps do not conform to body
Angles neck straps are cone/collar shaped	Neck strap edge may dig into neck muscles if poor fit
A scapula plate provides similar even pressure distribution as the breast plate	Neck straps bear weight over scapulae
Less stress and less rub potential over spine	Neck straps impede movement of shoulders
	Neck straps accelerate muscle fatigue
	Harness rub can cause localized hair loss / skin abrasion
X-Back	
Advantages	Disadvantages
Easy to put on and off	Side straps may hinder chest expansion
Difficult for dog to escape	Too short cross strips increase pressure at loin arch instead of tail base
Neck straps conform to curvature of body	Neck straps bear weight over scapulas
Optimal length of cross straps contribute pressure from loin arch to tail base	Neck straps may impede movement of shoulders
	Neck straps may accelerate muscle fatigue especially if poor fit
	Harness rub causes localized hair loss / skin abrasion
Collared neck (improved X-Back design)	
Improved movement of forelimb decreases pressure on scapula – scapula plate	More difficult to fit properly
X-straps around body caudal to ribcage increase ribcage expansion	If not fitted properly, can put pressure on trachea, neck or shoulders
Not fit tightly around the neck	If fit tight enough to gain advantages of this style, can be difficult to put on and remove
	Medium and heavy coated dogs may be uncomfortable during harnessing and unharnessing due to tight fit
	Harness rub can cause localized hair loss/ skin abrasion

Table 1: Advantages and disadvantages of various harness types.

The dog should be able to move and breathe comfortably when in harness. A correctly fitted harness will allow the dog to perform to its full potential, and prevent injuries. Each dog has a slightly different anatomy, and therefore each harness should be matched to the individual body. A perfectly fitted harness is not too tight, not too loose, and wraps snugly

Injuries to the Axial Skeleton of Sled Dogs

Dirsko J.F. von Pfeil¹, Dr.med.vet, DVM, DAVCS, DECVS; Caroline Griffiths², DVM

¹Veterinary Specialists of Alaska, P.C. 3330 Fairbanks Street, Anchorage, AK 99503, U.S.A; ²The Traveling Vet LLC, 6816 W C R 20, Loveland, CO 80537, U.S.A.

Orthopedic injuries of sled dogs can occur during training or racing, and frequently result in lameness. While there are numerous possible causes for lameness in the dog, this chapter will discuss clinical signs, diagnostic findings, treatment, and prognosis of the most commonly observed orthopedic injuries in the sled dog. A thorough knowledge of canine anatomy is essential to understand, diagnose, and treat orthopedic injuries. Interested readers are referred to the recommended literature on canine anatomy in the chapter on orthopedic examination of the sled dog in this book. That chapter also provides a glossary of terms for non-veterinarians.

Prevention of orthopedic injuries may be achieved by appropriate training methods and conditioning of the dogs. Although scarce, there is some literature available on the prevention of orthopedic injuries in the canine athlete. These suggestions are included in the beginning of this chapter. Some orthopedic injuries may be treated without surgery. Massage therapy or other non-invasive modalities may lead to a successful outcome. Several courses to learn these techniques are offered to people without a professional background in veterinary medicine, and to people who have been formally trained as veterinarians or veterinary technicians. Participants are provided with knowledge on canine anatomy, physiology, biomechanics, massage techniques, physical rehabilitation, behavior, handling and first aid.¹⁻² Other orthopedic injuries need to be treated surgically. The goal of this chapter is to help the reader understand how to prevent orthopedic injuries, which injuries may be treated on the trail, how appropriate treatment is performed, and which injuries may require surgical therapy.

PREVENTION OF INJURIES TO THE AXIAL SKELETON OF THE SLED DOG³

Selective breeding may help to prevent injuries from occurring. Although there is no proof yet, various orthopedic diseases or injuries may be based on a genetic predisposition. More and more attention is given to this in veterinary orthopedic research.⁴

Choosing the right equipment (neck lines, tug lines, tow line, harness, etc) may prevent injuries from occurring. Neck injuries may be prevented by the use of bungee cord necklines. These are elastic bands that will stretch out like a thick rubber band, indicating to the dog that it will soon reach the point of maximal pull. A similar device is frequently applied as the connection between tow line and sled. Some teams do not use necklines at all (Figure 1). This provides a larger range of motion of the neck and allows the dog to select its footing on the trail better. Not using neck lines requires the dogs to be well trained and disciplined, or bit wounds may occur.

The correct harness choice is also important. There are three main types of harness styles, the H-back, X-back, and Collared-Neck Harness (Figure 2). Each has its specific advantages and disadvantages (Table 1). These harnesses all consist of straps that go around the neck, meet in the middle of the chest between the front legs, and then extend over the lateral aspect of the chest towards the flank of the dog to finally end caudal to the base of the tail, where they are attached to the tug line. Several variations of these harness types can also be found.

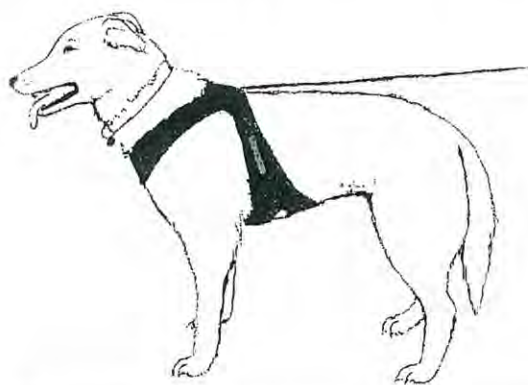


Figure 1: A short harness without neck lines provides more freedom for the dog and *may* decrease the risk for back injuries. Dogs using this type of harness need to be very disciplined.

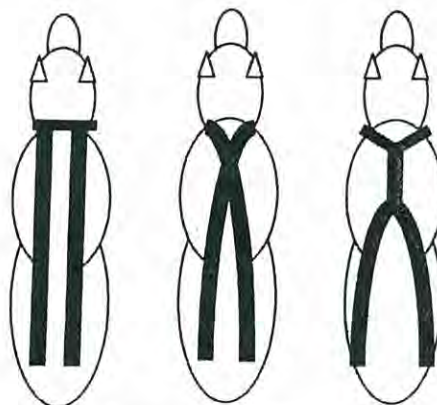


Figure 2: Commonly used harness types. H-Back, X-Back, Collared Neck (left to right).